



Machine Learning Operations (MLOps) AIMLCZG523

Assignment 01

FAQs

Q. Which dataset should be used?

The assignment requires you to use the **Heart Disease UCI Dataset** from the UCI Machine Learning Repository.

Q. What kind of Exploratory Data Analysis (EDA) is expected?

You should include:

- Histograms
- Correlation heatmaps
- Class distribution plots
- Missing value analysis
- Feature relationship analysis

Professional visualizations using Matplotlib, Seaborn, or Plotly are recommended.

Q. Should we clean and preprocess the data?

Yes. You should:

- Handle missing values
- Encode categorical variables
- Scale numerical features (if applicable)
- Split train/test data properly

Q. How many models should be trained?

At least two classification models are expected.

Examples:

- Logistic Regression
- Random Forest
- XGBoost
- Support Vector Machine or any choice of yours

Q. Which evaluation metrics should be used?

Recommended metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Cross-validation is encouraged.

Q. Do we need hyperparameter tuning?

Yes. Basic tuning using:

- GridSearchCV
- RandomizedSearchCV
- Optuna (optional) or any other

is recommended.

Q. Why are we using MLflow?

MLflow helps track:

- Experiments
- Parameters
- Metrics
- Model versions
- Artifacts

This ensures reproducibility and proper experiment management.

Q. What should be logged in MLflow?

You should log:

- Model parameters
- Accuracy metrics
- ROC curves/plots
- Confusion matrix
- Saved models

Q. Why do we need requirements.txt?

It allows the project to run in a clean environment with all required dependencies installed consistently.

Q. Should preprocessing steps also be saved?

Yes. Preprocessing pipelines must be reusable during inference.

Recommended approach:

- sklearn Pipeline
- ColumnTransformer

Q. Which model serialization format can be used?

Any of the following are acceptable:

- Pickle
- Joblib
- MLflow model format
- ONNX

Q. Which framework should be used for the API?

Recommended:

- FastAPI (preferred)
- Flask

Q. What should the API expose?

A /predict endpoint that:

- Accepts JSON input
- Returns prediction
- Returns confidence/probability score

Q. How can the API be tested?

You may use:

- Postman
- curl
- Swagger UI (FastAPI)
- pytest

Q. Why are we using Docker?

Docker ensures the application runs consistently across environments by packaging:

- Code
- Dependencies
- Runtime environment

inside a container.

Q. What should be included in the Docker container?

The container should include:

- API code
- Trained model
- Dependencies
- Inference pipeline

Q. How do we verify Docker works? You should:

1. Build the Docker image
2. Run the container locally
3. Test the /predict endpoint successfully

Q. What is expected in the CI/CD pipeline?

The pipeline should automate:

- Code linting
- Unit testing
- Model training (optional)
- Build validation

Q. Which CI/CD tools are allowed?

You may use:

- GitHub Actions (recommended)
- Jenkins
- GitLab CI/CD

Q. Should the pipeline fail if tests fail?

Yes. The pipeline must stop and clearly report errors if:

- Tests fail
- Code quality checks fail

Q. Is cloud deployment mandatory?

You may deploy using:

- Public cloud (AWS/GCP/Azure)
- OR
- Local Kubernetes (Minikube/Docker Desktop Kubernetes)

Q. Which Kubernetes platforms are acceptable? Examples:

- Minikube
- Docker Desktop Kubernetes
- Amazon EKS
- Google GKE
- Azure AKS

Q. What Kubernetes files are expected?

You should provide:

- Deployment YAML
- Service YAML
- Optional Helm chart

Q. What monitoring is expected?

Basic monitoring such as:

- API request logging
- Response monitoring
- Prometheus metrics
- Grafana dashboards

is sufficient.

Q. Why is monitoring important in ML systems?

Monitoring helps detect:

- Model failures
- Data drift
- API downtime
- Performance degradation

Q. What should the final report contain?

The report should include:

- Project overview
- EDA findings
- Model comparison
- Architecture diagram
- MLflow screenshots
- CI/CD screenshots
- Deployment screenshots
- Setup instructions
- Repository link



Q. Is a GitHub repository mandatory?

Yes. A clean and properly organized repository is expected.

Recommended Tool Stack

Category	Recommended Tools
Programming Language	Python
Notebook Environment	Jupyter Notebook, VS Code
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
ML Frameworks	Scikit-learn, XGBoost
Experiment Tracking	MLflow
API Framework	FastAPI, Flask
Testing	Pytest
Containerization	Docker
CI/CD	GitHub Actions, Jenkins
Orchestration	Kubernetes, Minikube
Monitoring	Prometheus, Grafana
Version Control	Git, GitHub
Deployment Platforms	AWS, GCP, Azure, Render, Railway